

CAF Current State — Vol 2: Data & Contracts

Aggregated export — 2026-07-10. Source markdown in repo is unchanged.

Bundle id: 12-caf-current-state-vol2-data

Included:

docs/volumes/CAF_CONTEXT_VOL2_Data_Contracts_and_Flows.n

CAF Current State Context Pack — Data Contracts and Flows

Volume 2 of 4 | Generated 2026-07-10 | Full pack: docs/CAF_CURRENT_STATE_CONTEXT_PACK.md

5. Database schema map

Grouped by domain (79 migrations through 078_brand_bibles.sql):

Core pipeline

projects , runs , content_jobs , job_drafts , assets , job_state_transitions ,
signal_packs , signal_pack_ideas , signal_pack_selected_ideas , ideas ,
idea_grounding_insights

Writers: run-orchestrator.ts , job-pipeline.ts , runs.ts

Readers: all routes keyed by task_id

Review / QC / validation

editorial_reviews , diagnostic_audits , auto_validation_results ,
qc_checklists , qc_flow_profiles

Writers: validation-router.ts , review routes, qc-runtime.ts (payload only)

Flow engine / prompts

flow_definitions , prompts , prompt_versions , prompt_schemas ,
carousel_templates , risk_policies

Writers: migrations seed, flow-engine.ts admin

Readers: llm-generator.ts , qc-runtime.ts

Project configuration

project_strategy , brand_constraints , product_profiles ,
platform_constraints , project_system_constraints , project_brand_assets ,
brand_profiles , brand_bibles , heygen_config , project_integrations ,
risk_rules (not QC-enforced)

Inputs / evidence

`inputs_evidence_imports` , `inputs_evidence_rows` ,
`inputs_evidence_row_insights` , `inputs_evidence_packs` ,
`inputs_processing_profiles` , `inputs_source_rows` , `inputs_scraper_config` ,
`inputs_scraper_runs` , `inputs_idea_lists` , `inputs_ideas` ,
`evidence_media_assets` , `insights_packs`

Creative intelligence / mimic

`creative_intelligence_assets` , `creative_intelligence_analyses` ,
`creative_intelligence_insights` (migration 055+)

Learning / performance

`learning_rules` , `suppression_rules` , `learning_observations` ,
`learning_hypotheses` , `learning_insights` , `learning_generation_attribution` ,
`performance_metrics` , `performance_ingestion_batches` , `run_content_outcomes` ,
`job_outcomes` (075), `llm_approval_reviews`

Publishing

`publication_placements` , `project_integrations`

Operations / audit

`api_call_audit` , `run_context_snapshots` , `run_output_reviews`

6. Lifecycle map

Runs

See §3. Terminal: `COMPLETED` , `FAILED` , `CANCELLED` .

Content jobs

See §3. Editorial terminal: `APPROVED` , `REJECTED` . Rework re-enters `GENERATING` .

Editorial decisions (`editorial_reviews.decision`)

`APPROVED` | `NEEDS_EDIT` | `REJECTED`

Publication placements

`draft` → `scheduled` → `publishing` → `published` | `failed` | `cancelled`

Learning rules

`pending` → `active` | `superseded` | `rejected` | `expired`

Provider render state (`generation_payload.render_state`)

HeyGen: `video_id` , `session_id` — retry guard via `hasActiveProviderSession()`
(`content-job-render-state.ts`).

Scene assembly: `scene_bundle` progress, Sora job IDs.

Carousel: `isCarouselRenderComplete()` .

QC recommended routes (`qc_result.recommended_route`)

AUTO_PUBLISH | BLOCKED | DISCARD | REWORK_REQUIRED | HUMAN_REVIEW
Default CAF_REQUIRE_HUMAN_REVIEW_AFTER_QC=true remaps clean QC to HUMAN_REVIEW .

7. Critical JSON contracts

content_jobs.generation_payload (main integration contract)

Slice	Meaning	Read/write helpers
Plan-time base	schema_version, signal_pack_id, candidate_data, prompt binding	buildPlannedGenerationPayloadBase() (stage-contract.ts)
generated_output	LLM output body	pickGeneratedOutput(), hasGeneratedOutput()
qc_result	QC pass/fail, findings, route	pickStoredQcResult(), mergeGenerationPayloadQc() (only writer)
render_state	Provider sessions, carousel progress	pickRenderState(), hasActiveProviderSession(), isMidProviderPhase()
mimic_v1	Mimic render truth	pickMimicPayload(), MIMIC_PAYLOAD_KEY
mimic_carousel_package	Draft package for TP-grounded carousels	pickMimicCarouselDraftPackage() — all isTpGroundedCarouselRenderFlow()
carousel_package	Standard FLOW_CAROUSEL only	draft-package-contract.ts
bvs_v1	Frozen brand bible snapshot	parseBvsV1(), attachBvsToPlannedPayload()
mimic_job_grounding	Plan-time mimic grounding	mimic-job-grounding.ts
semantic_contract_v1	Cross-flow semantic constraints	semantic-contract.ts
content_display	Display metadata for Review	content-display-metadata.ts
layout_qc	Post-render layout QA blob	mimic-composite-layout-qa.ts
docai_layer_positions	Inside mimic_v1 — reviewer text boxes	mimic-docai-layer-positions.ts
scene_bundle	Scene assembly progress	scene-pipeline types
schema_validation_warnings	Non-fatal LLM schema issues	when CAF_OUTPUT_SCHEMA_VALIDATION_MODE=warn
Publish snapshots	URLs after approval	validation-router.ts

mimic_v1 key fields

```
// src/domain/mimic-payload.ts (abridged)
{
  schema_version: 1,
  execution_mode?: "classic" | "why_mimic" | "new_visual",
  mode: "image_full" | "template_bg" | "carousel_visual",
  reference_items[], visual_guideline, slide_plans[],
  flux_image_prompts?, docai_layer_positions?,
  bvs_enabled?, bvs_bible_snapshot?, bvs_render_plan?,
```

```
    slide_intelligence?, brand_execution_brief?
  }
```

bvs_v1 key fields

```
// src/domain/bvs-v1.ts
{ schema_version: "bvs_v1", enabled: boolean, bible_version, bible_snapshot }
```

Signal pack JSON (high-signal fields)

- `overall_candidates_json` — raw idea rows
- `visual_guidelines_pack_v1` — merged styling for mimic
- `mimic_mode_overrides` — per-insight reviewer mode overrides
- `derived_globals_json` — CI styling merge output

Risky untyped areas

- Ad-hoc reads of `candidate_data` nested fields in Review components
- Legacy `ideas_json` on older signal packs
- `render_manifest` (commented in migrations, sparse usage)
- Some admin HTML forms bypass typed helpers

8. Flow types and content formats

Canonical text/copy flows (`canonical-flow-types.ts`)

`FLOW_CAROUSEL` , `FLOW_ANGLE` , `FLOW_STRUCTURE` , `FLOW_CTA` , `FLOW_HOOKS` ,
`FLOW_TEXT`

Video flows

Flow	Status	Generate	Render	Review
<code>FLOW_VID_SCRIPT</code>	Production	OpenAI script	HeyGen	Video edits panel
<code>FLOW_VID_PROMPT</code>	Production	OpenAI prompt	HeyGen Video Agent	HeyGenReviewEdits
<code>FLOW_VID_PROMPT_NO_AVATAR</code>	Production	OpenAI	HeyGen no-avatar	HeyGenReviewEdits
<code>FLOW_VID_SCENES</code> / <code>FLOW_SCENE_ASSEMBLY</code>	Provider-dependent	Scene scripts	Sora clips + ffmpeg concat	Video review
<code>FLOW_PRODUCT_*</code> (6 types)	Production	Product prompts	HeyGen	Video review

Carousel flows

Flow	Status	Notes
<code>FLOW_CAROUSEL</code>	Production	<code>carousel_package</code> , HBS templates, <code>CarouselBrandStylingPanel</code>

FLOW_TOP_PERFORMER_MIMIC_CAROUSEL	Production (gated render)	mimic_v1, reference frames, compare UI in Review
FLOW_VISUAL_FIRST_CAROUSEL	Production (evolving)	execution_mode: "new_visual" — no TP references, BVS-driven
FLOW_WHY_MIMIC_CAROUSEL	Production (newer)	SIL-driven, execution_mode: "why_mimic"
FLOW_TOP_PERFORMER_MIMIC_IMAGE	Gated	image_full mode, single frame
FLOW_TOP_PERFORMER_MIMIC_VIDEO	Alias	Routes to HeyGen video planning

Product image flows (FLOW_IMG_*)

Registered in flow engine; **blocked at LLM** with PRODUCT_IMAGE_FLOW_NOT_READY_MESSAGE .

Offline / excluded (offline-flow-types.ts)

Reel_Script , FLOW_REEL_SCRIPT , hook variation flows — excluded from planning.

Per-flow requirements

Flow family	Env / config
Mimic carousel	MIMIC_IMAGE_ENABLED=1, provider API key, archived reference media (except new_visual)
New visual	BVS recommended; analysis_t2i input mode
HeyGen	HEYGEN_API_KEY, project heygen_config
Scene assembly	OPENAI_API_KEY, SCENE_ASSEMBLY_CLIP_PROVIDER, VIDEO_ASSEMBLY_BASE_URL
Standard carousel	RENDERER_BASE_URL, templates
